

Ambiguity

- RDCQA: property values without property names
- "2007": year, dollar amount, number of people, aggregated value?
- domain-independent keyphrase preprocessing
- match combination
- target dataset ambiguity minimized through whole-query scoring



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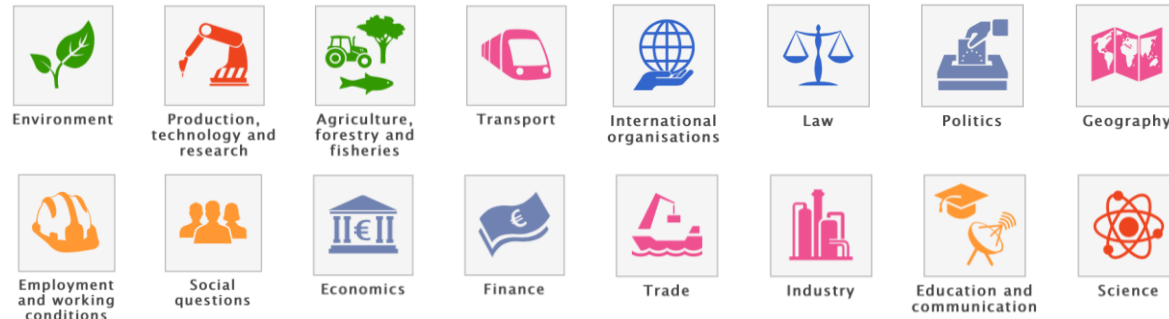
Multi-level semantic labelling of numerical values

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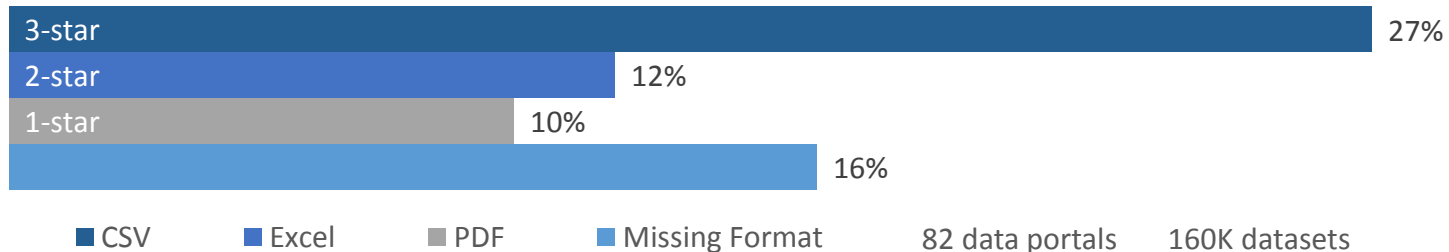
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Motivation: Open Data

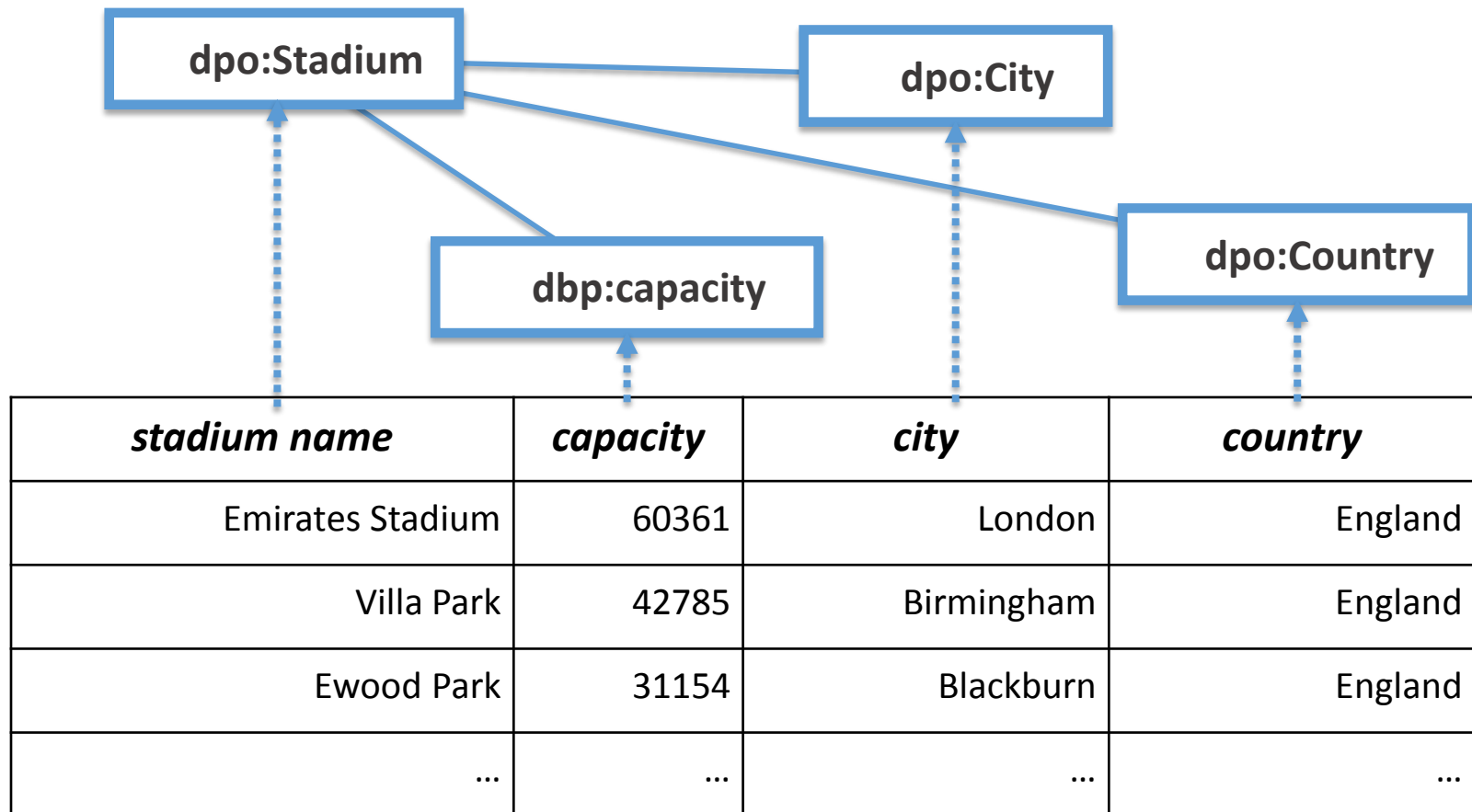


- Available information is (partially) structured and tabular [1]:



- Exploiting the potential:
 - Improve existing 3-star (CSV) Open Data
 - Propose semantic labels/context for the content of columns
 - Integrate and interlink data

Example



But:

Web/HTML tables differ from typical Open Data tables:

- **Domain:** e.g., public administration data, statistical data, weather data, elections, ...
- **Structure:** OD tables contain large amount of numerical columns

Wohnungen in den 250 Zaehlbezirken in Wien - Registerzaehlung 2011 Housing units in 250 su					
NUTS1	NUTS2	NUTS3	DISTRICT_CODE	SUB_DISTRICT_CODE	WHG_TOTAL
AT1	AT13	AT130	90100	90101	3004
AT1	AT13	AT130	90100	90102	1049
AT1	AT13	AT130	90100	90103	1389
AT1	AT13	AT130	90100	90104	1014
AT1	AT13	AT130	90100	90105	1337
AT1	AT13	AT130	90100	90106	1915
AT1	AT13	AT130	90100	90107	2032
AT1	AT13	AT130	90200	90201	5178
AT1	AT13	AT130	90200	90202	6345
AT1	AT13	AT130	90200	90203	7549
AT1	AT13	AT130	90200	90204	8388
AT1	AT13	AT130	90200	90205	5358
AT1	AT13	AT130	90200	90206	4237
AT1	AT13	AT130	90200	90207	7812
AT1	AT13	AT130	90200	90208	1478
AT1	AT13	AT130	90200	90209	7547

Example (Cont'd)

<i>stadium</i>	<i>capacity</i>	<i>city</i>	<i>country</i>
Emirates Stadium	60361	London	England
Villa Park	42785	Birmingham	England
Ewood Park	31154	Blackburn	England
...	...	...	...

Example (Cont'd)

	<i>TOTAL</i>	<i>DISTRICT_CODE</i>	<i>ISO_2</i>
Emirates Stadium	60361	SW1A 0AA	GB
Villa Park	42785	B23 7QG	GB
Ewood Park	31154	B26 6QA	GB
...	...	...	...

Why not use numeric values?

- Identifying the most likely semantic label for a bag of numerical values
- Deliberately ignore surroundings

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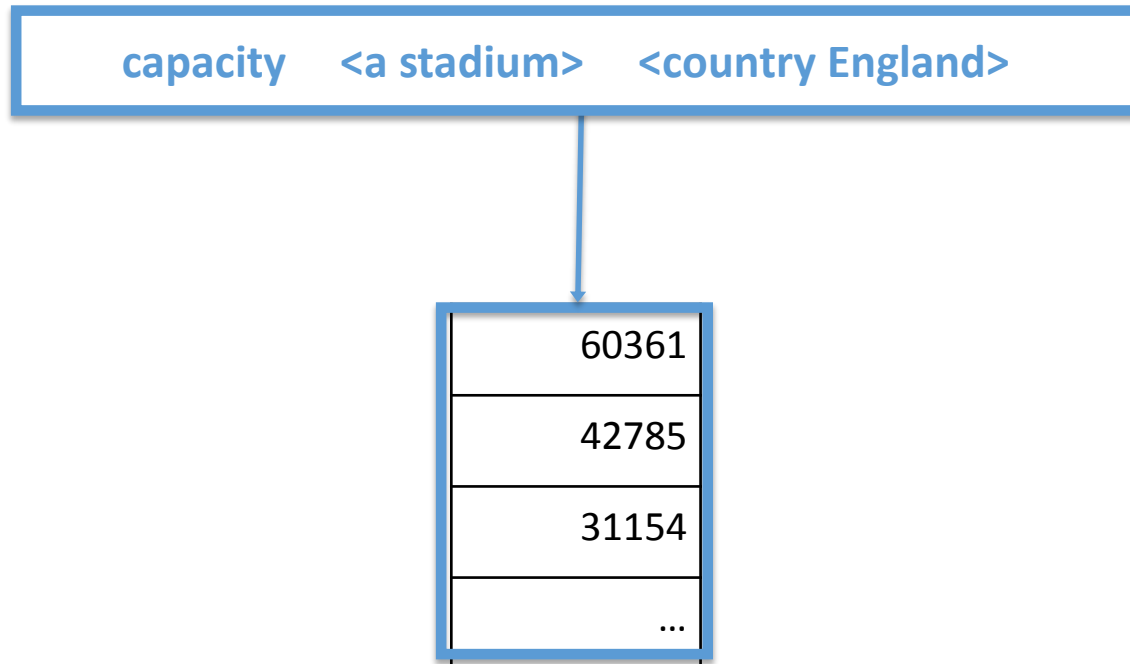
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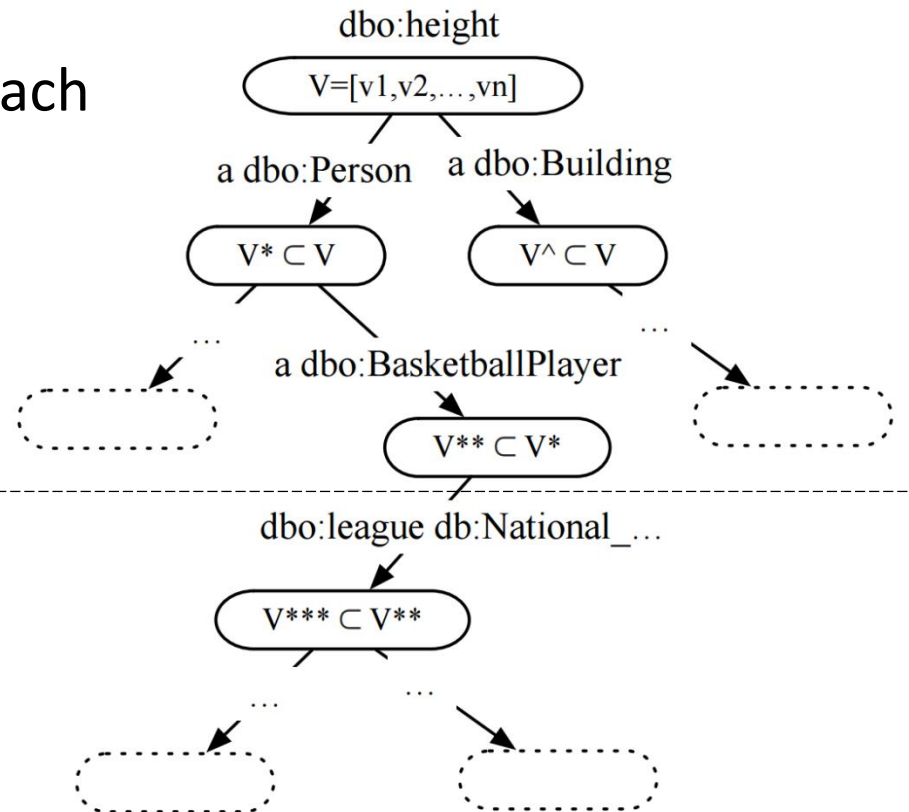


Our Approach

- 1. Hierarchical clustering** over an RDF knowledge base
 - to build background knowledge graph (**BKG**)
 - nodes consist of **typical numerical values**, annotated with context information, i.e.:
grouped by **properties** and their **shared domain (subject) pairs**
- 2. k-nearest neighbors search**
- 3. Aggregation of the results** at different levels to find the most likely context:
 - property
 - type
 - context

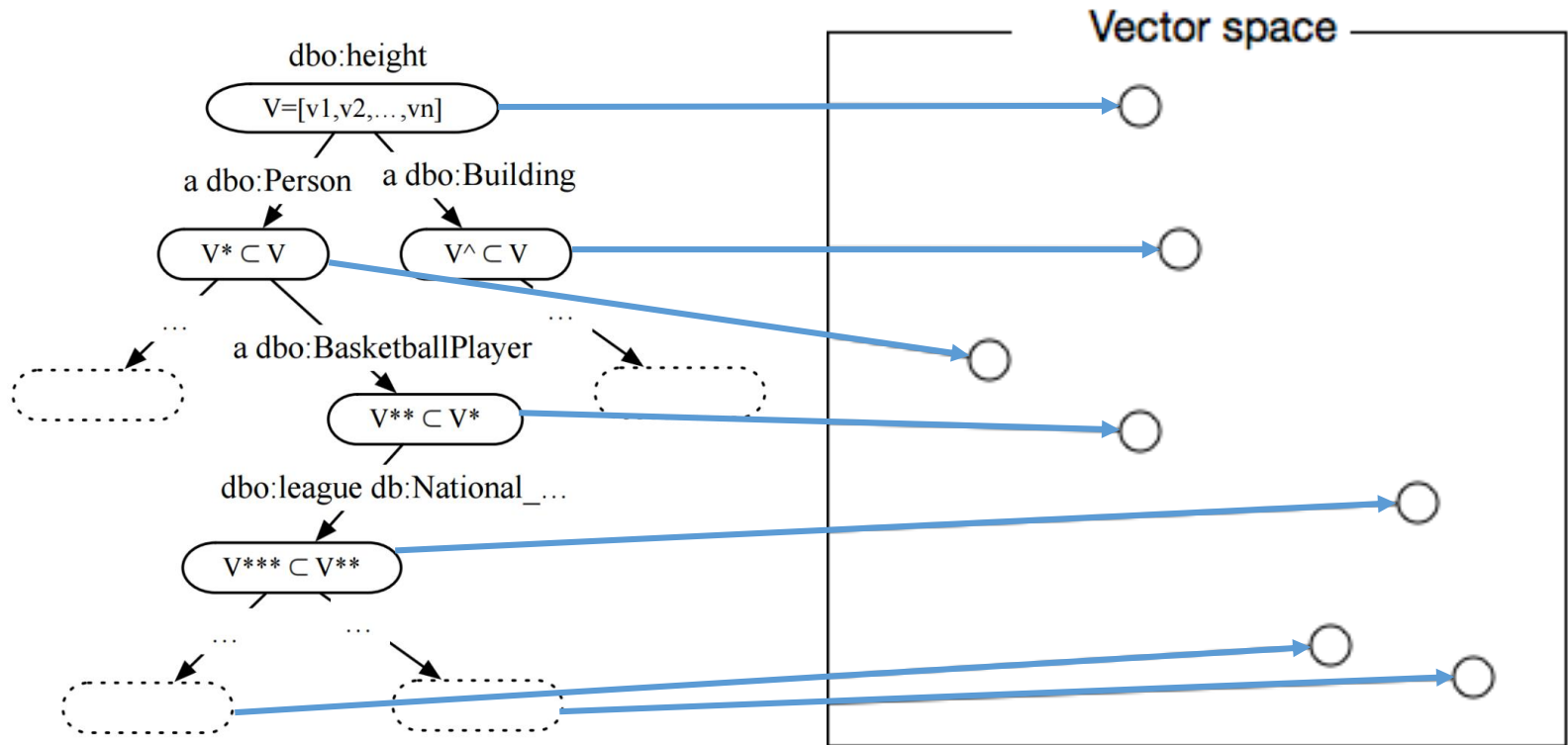
1. Background Knowledge Graph

- Find properties with **numerical range**
- Hierarchical clustering approach
- Two hierarchical layers:
 - **Type** hierarchy (using OWL classes)
 - **Property-object** hierarchy (shared property-object pairs)



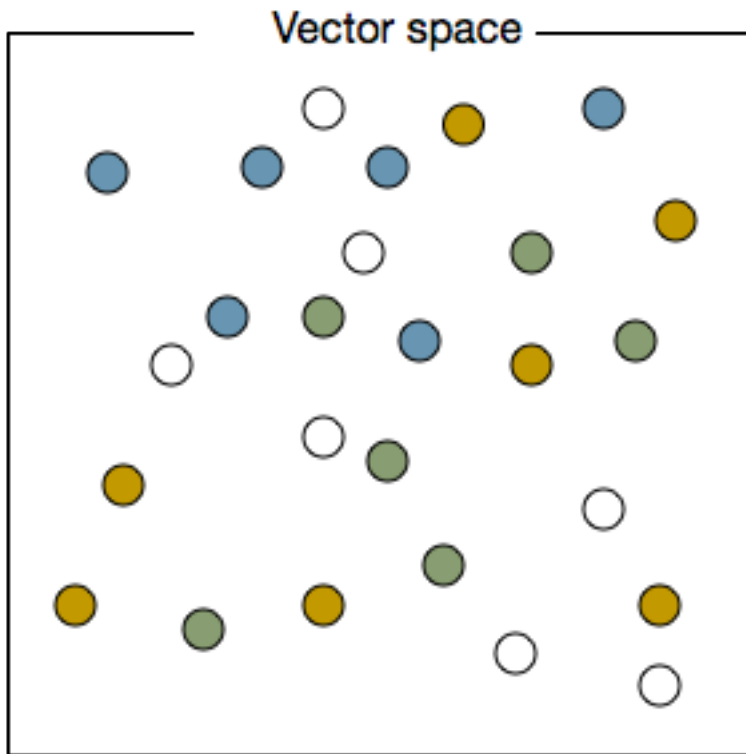
2. *k*-Nearest neighbor search

Mapping bags of numerical value to vector space (feature vector)



2. *k*-Nearest neighbor search (cont'd)

Compute & rank *k*-nearest neighbours for input values



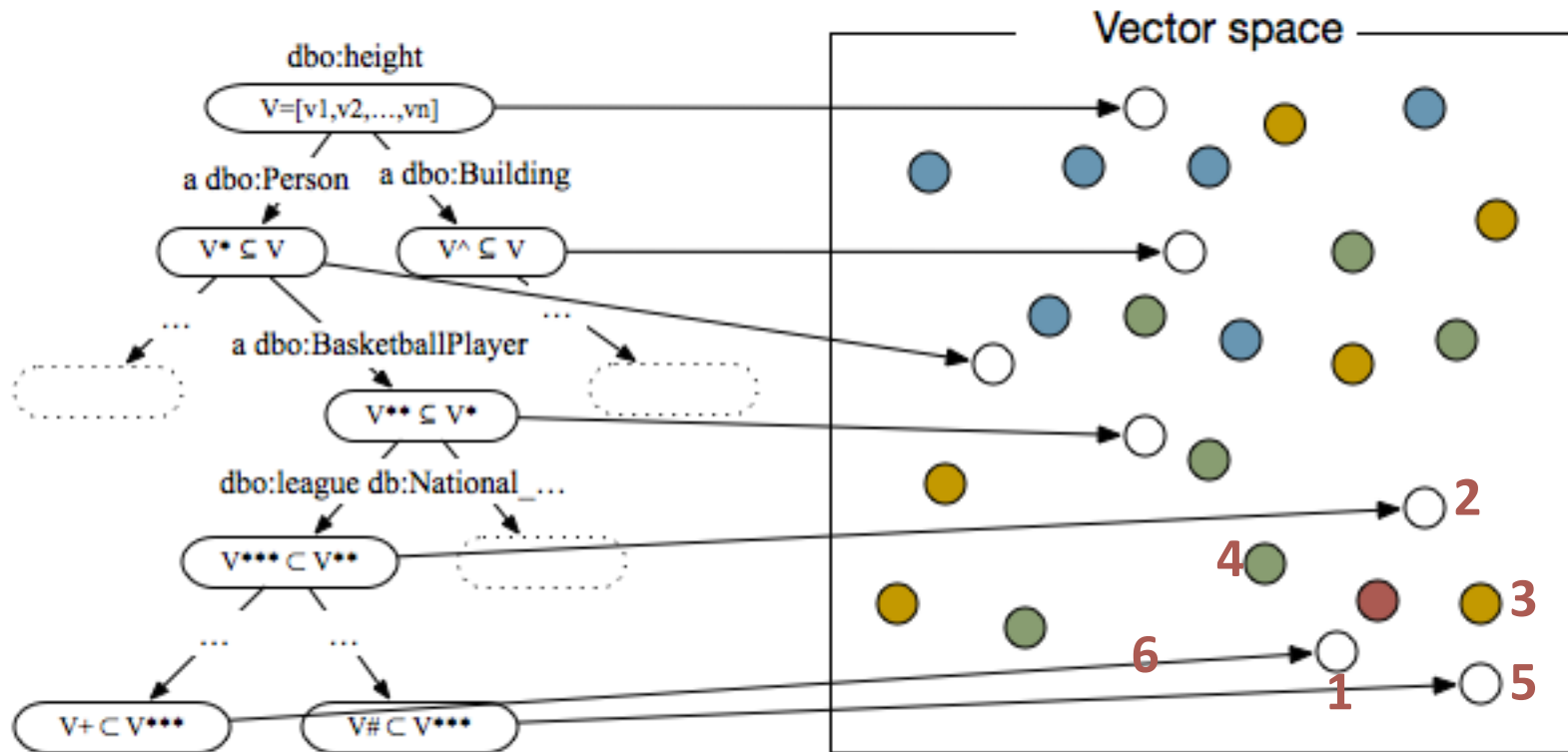
1) input: [187, 201, 199, 198, 195, 199, 203, ...]

2) mapping: 

3) compute distance to neighbours

4) select *K* nearest

3. Result Aggregation



Evaluation: Setup

- Data
 - DBPedia 3.9
 - 50 most frequent numerical properties
- Aggregation Levels
 - *property*
 - *type*
 - *p-o level*
- Aggregation Function
 - *majority vote* and *average distance*
- Evaluation of different distance functions
 - Best: *Kolmogorov-Smirnov (KS) distance*

Evaluation

33657 Test Nodes

- Majority vote slightly better
- More neighbors also better
- Top-5 shows already clear better results than top-1
- 9% of test nodes are contained 1-1 in knowledge graph !!

Accuracy results:

neigh.	top- <i>k</i>	prop		type		p-o	level
	agg. results	maj.	avg.	maj.	avg.	maj.	avg.
25	1	59.3	34.5	64.7	57.8	20.4	24.9
	5	87.9	82.9	91.4	85.3	75.8	66.2
	10	98.5	98.5	94.7	94.7	83.8	74.0
50	1	57.4	23.7	66.4	37.6	20.4	24.9
	5	98.4	83.7	93.2	65.4	75.8	66.2
	10	99.3	99.3	96.3	96.1	83.8	74.0

Experimental OD Column labelling

- Data from two selected Open Data portals
 - 1170 CSV tables
- Manual inspection of top 100 tables
- **Lessons learned:**
 - Missing domain knowledge
 - Timeline data
 - Combine with (existing) complementary approaches

Conclusions

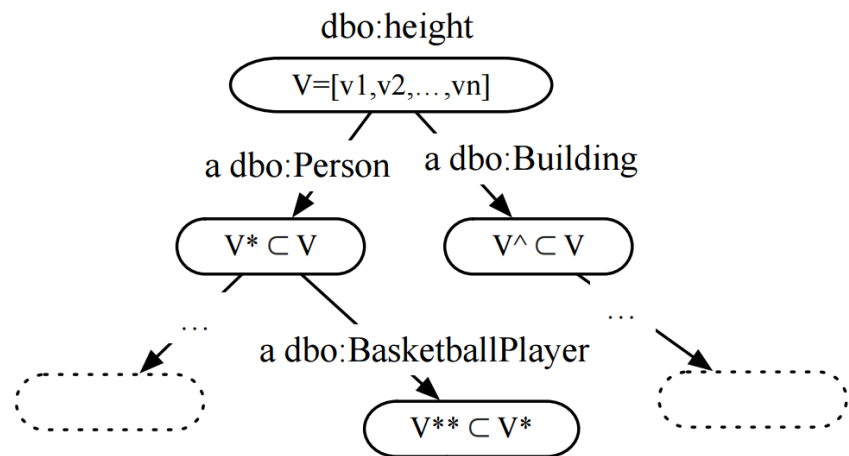
- Semantic labelling of numerical values
 - Hierarchical unsupervised background knowledge (BK)
- We can assign fine-grained semantic labels if there is enough evidence in BK
- Complementary to existing approaches
- Future work
 - Find and integrate more background knowledge
 - Solve the domain mismatch between Open Data and existing KGs
 - Potentially applicable in other use cases



Backup Slides

BKG: Type hierarchy

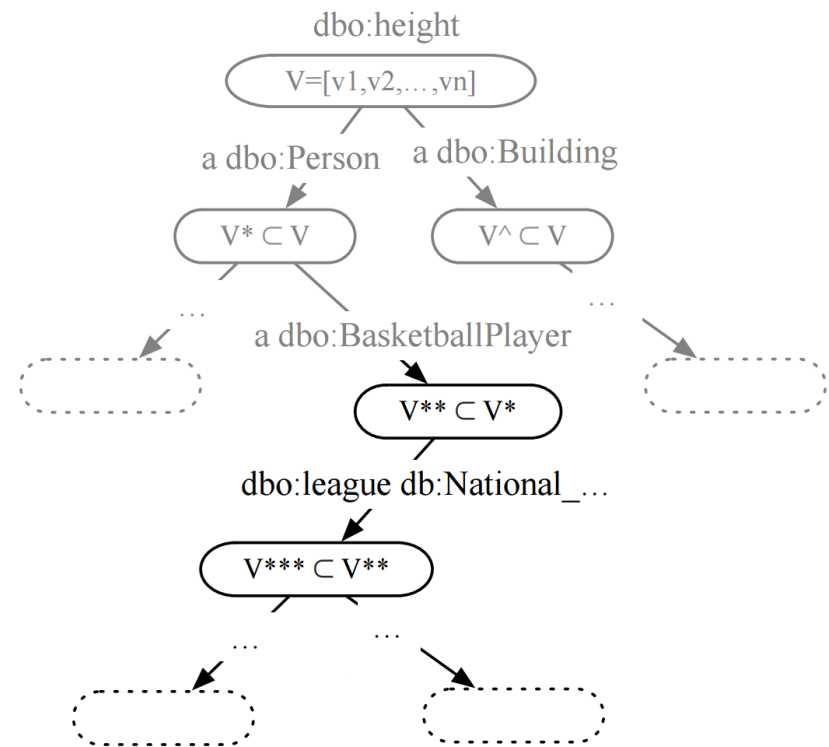
- Represents `rdfs:subClassOf` relation for all available types
- Construction:
 1. Find properties with **numerical range**
 2. Collect **entities** and all of their **p-o pairs**
 3. Materials the OWL class hierarchy
 4. Form a cluster for each type, containing all entity information



BKG: p-o hierarchy

- Hierarchical clustering approach

1. Start with one node cluster
2. Build candidates:
 - **constrain property-object**: all subject share the same property-object pair
 - **constrain size**: candidate nodes are larger than 1% of the parent node and smaller than 99%
3. Sort candidates by their distance
4. Select candidate with largest distance, subsequently select non-overlapping candidates



Evaluation: Setup

- 30 GB RAM
- 3 different knowledge bases:

ID	DISTANCE MEASURE	NODES	BUILD TIME	AVG. PRED. TIME
KS	Kolmogorov-Smirnov test	11431	30m	2.5s
FV1	(min, max, mean, std)	11432	24m	2.3s
FV2	(5-q, 95-q, mean, std)	11432	38m	4.6s

- DBpedia properties:

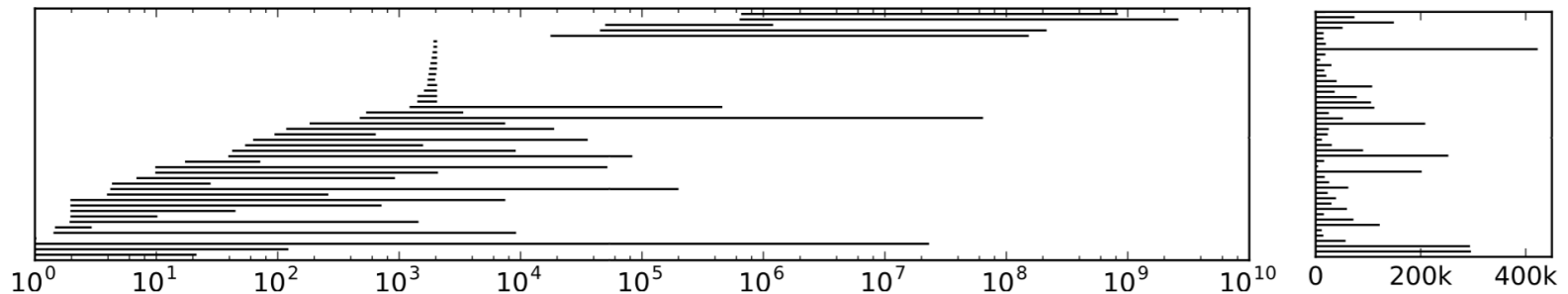


Fig. 4. 5%-95% inter-quantile ranges and number of values of train properties.

Evaluation: Test/train data

- train/test split : 80/20
- 20% of the subjects for each property as test data
- test context graph: similar as background construction, however, without constrains
- randomly select leaf nodes

Evaluation: Distance Function

1787 Test Nodes

	FV1			FV2			KS		
	Top-1	Top-5	Top-10	Top-1	Top-5	Top-10	Top-1	Top-5	Top-10
exact	2.5	8.2	8.2	2.5	8.2	8.2	12.3	41.8	47.9
prop	45.4	60.3	60.3	45.4	60.3	60.3	57.1	74.1	79.8
type	11.3	24.9	24.9	11.3	24.9	24.9	16.1	43.9	56.0
stype	24.9	41.1	41.1	24.9	41.1	41.1	35.8	58.6	67.5

Table 3. Accuracy in % for different distance functions

- Best:
Kolmogorov-Smirnov (KS) distance

exact = correct property, type and p-o
prop = correct property
type = correct type
stype = correct super type

Open Data (OD) tables

Open Data tables typically contain

- a large portion of numerical columns
- missing headers
- and/or non-textual headers

BabelNet



Portal	Tables	$\overline{cols}$	$\overline{num.cols}$	w/o Header	Num. H.	Mapped
AT	968	13	8	154	6,482	1,323
EU	357	20	4	223	1,233	349

Table 5. Header mapping of CSVs in Open Data portals

solutions that solely focus on textual “cues” are only partially applicable for mapping such data sources

Lessons Learned from OD tables

- Missing domain knowledge
 - Open Data is potentially very domain specific
 - Mismatch between knowledge bases like DBpedia and Open Data
 - Enrich knowledge graph with domain knowledge (e.g., extracted from Open Data tables)
- Timeline data
 - values for different time points not in DBpedia
 - Detect time dependency and regroup/transform tables
- Include complementary approaches
 - Deliberately excluded in this paper
 - Linguistic clues and string similarity measures (cf. approach by Pham et al.)